

L2 – Trig Ratios for Angles Greater than 90°

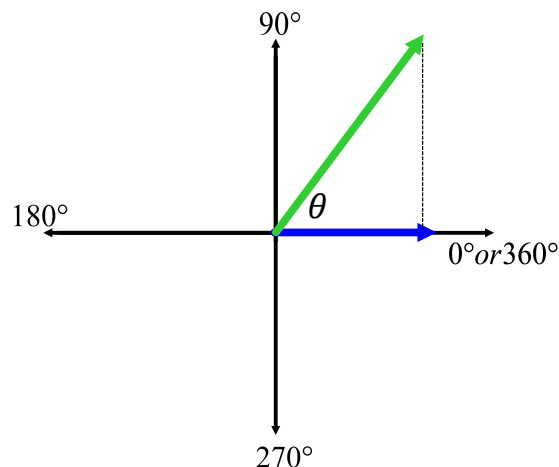
MCR3U

Part 1: Reference Angles

_____ : Always lies on the positive x -axis at 0° . Meets the terminal arm at the origin.

_____ : The arm that rotates around the origin counter clockwise to form a positive angle or clockwise for a negative angle.

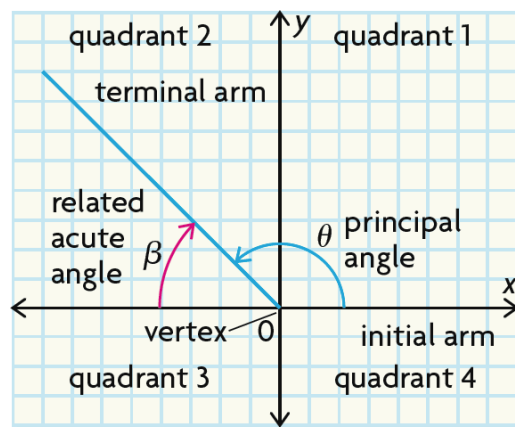
_____ is measured from the initial arm to the terminal arm.



_____ : The counter clockwise angle between the initial arm and the terminal arm of an angle in standard position. It's value is between 0° and 360° .

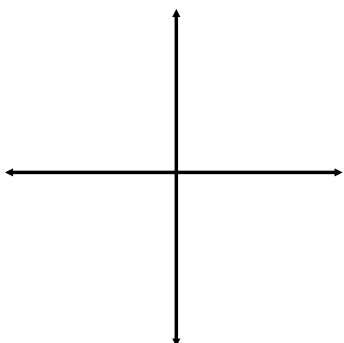
_____ : The acute angle between the terminal arm of an angle in standard position and the closest x -axis when the terminal arm lies in quadrant 2, 3, or 4.

The reference angle helps us determine the exact trig ratios when we are given obtuse angles.

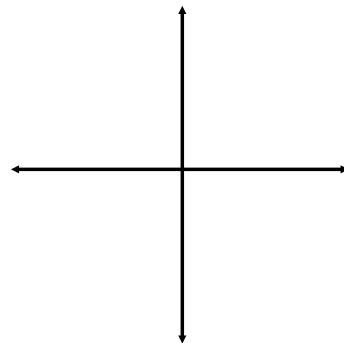


Example 1: Find the reference angle for each of the following principal angles

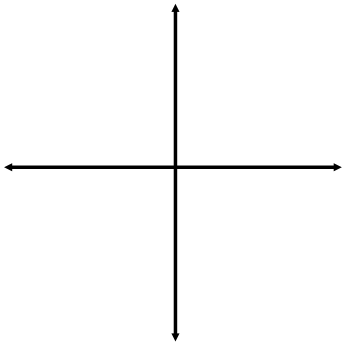
a) 250°



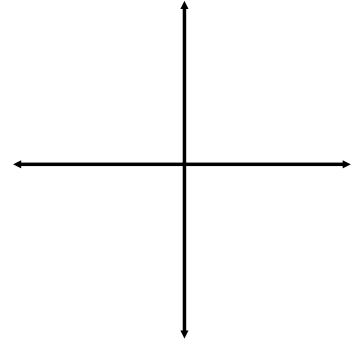
b) 120°



c) 300°



d) 225°



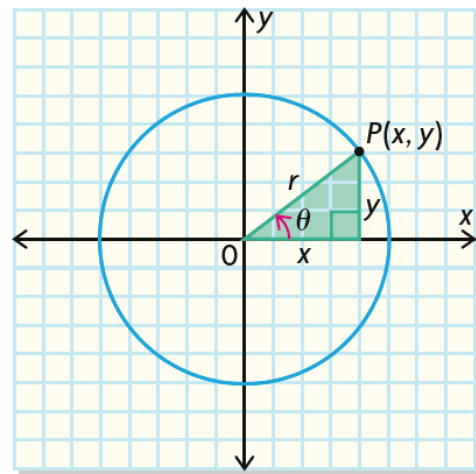
Part 2: Evaluating Trig Ratios for Any Angle

For any point $P(x, y)$ in the Cartesian plane, the trigonometric ratios for angles in standard position can be expressed in terms of x , y , and r .

$$\sin \theta =$$

$$\cos \theta =$$

$$\tan \theta =$$



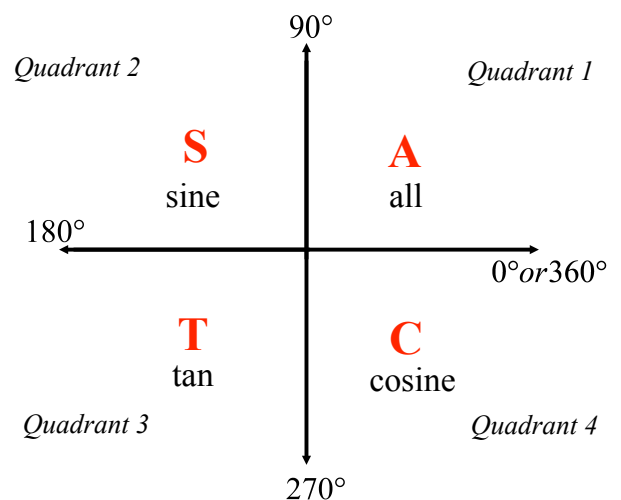
The CAST rule is an easy way to remember which primary trig ratios are positive in which quadrant. Since r is always positive, the sign of each primary ratio depends on the signs of the coordinates of the point (x, y) .

In Q1, _____ ratios are positive because both x and y are positive.

In Q2, only _____ is positive, since x is negative and y is positive.

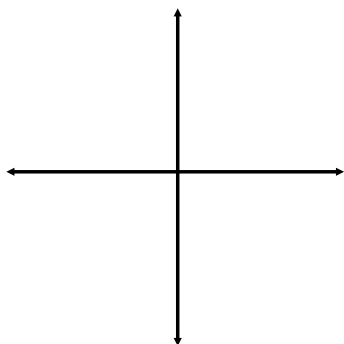
In Q3, only _____ is positive, since both x and y are negative.

In Q4, only _____ is positive, since x is positive but y is negative.

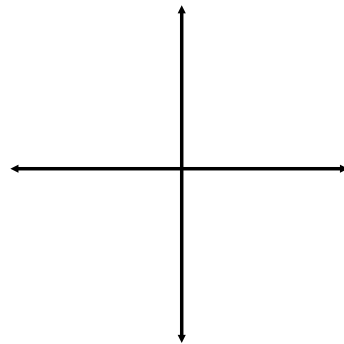


Example 2: Find the EXACT value of each of the following

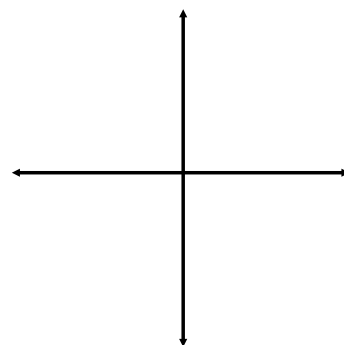
a) $\sin 45^\circ$



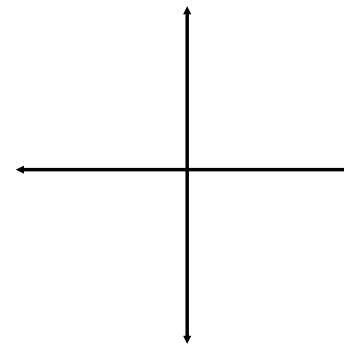
b) $\sin 210^\circ$



c) $\cos 240^\circ$

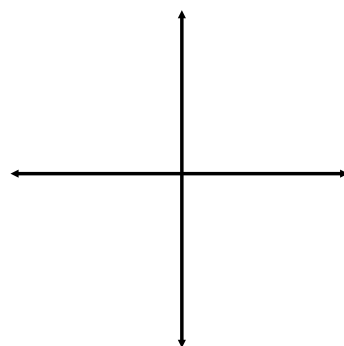


d) $\tan 315^\circ$

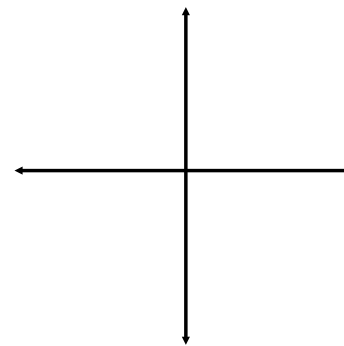


Example 3: Each point lies on the terminal arm of angle θ in standard position. Determine each of the primary trig ratios for angle θ .

a) $(5, -12)$



b) $(-8, 3)$



Part 3: Unit Circle

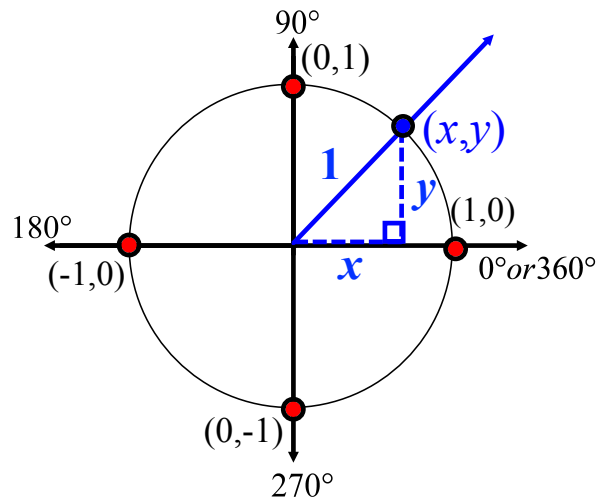
The unit circle, a circle with a radius of **1 unit**, is very useful since the x and y coordinates of where the terminal intersects it tell us the Cosine and Sine ratios respectively.

For any point $P(x, y)$ in the Cartesian plane that intersects the **unit circle**, the trigonometric ratios for angles can be expressed in terms of x , y , and r .

$$\sin \theta =$$

$$\cos \theta =$$

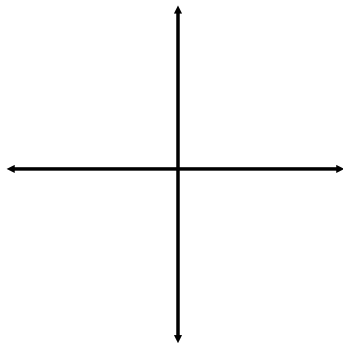
$$\tan \theta =$$



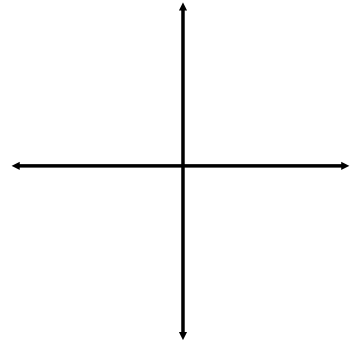
<http://www.mathsisfun.com/geometry/unit-circle.html>

Example 4: Find the EXACT value of each of the following

a) $\sin 270^\circ$



b) $\cos 360^\circ$



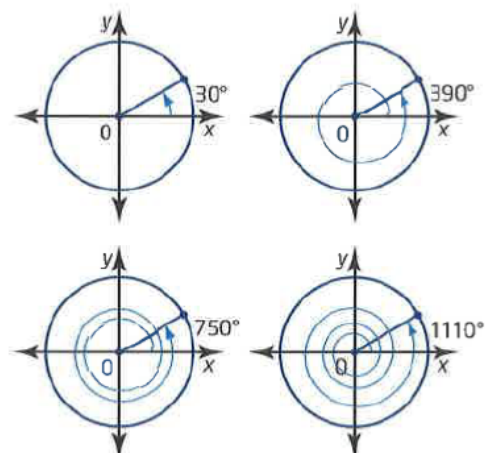
Part 4: Negative and Co-terminal Angles

Co-terminal angles are angles in standard position that have the _____.

Starting at 30° and rotating 360° counter clockwise will bring you back to the same terminal arm.

$$30^\circ + 360^\circ = 390^\circ$$

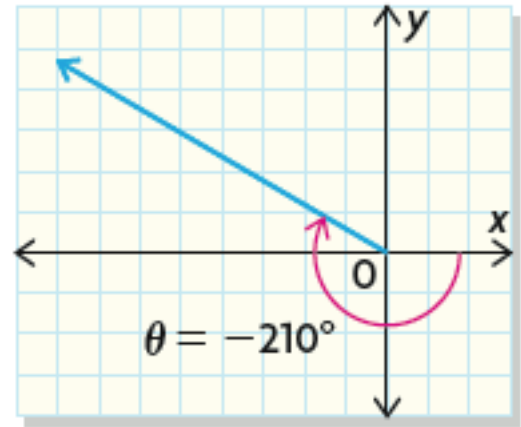
Therefore, 30° and 390° are co-terminal.



A negative angle is an angle measured _____ from the positive x -axis.

You can find an equivalent (co-terminal) positive angle by adding 360° to the negative angle.

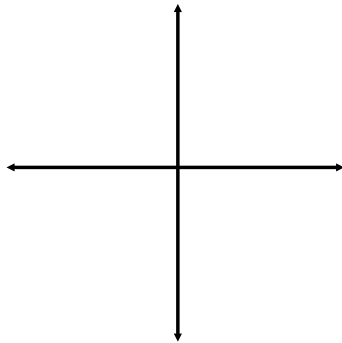
-210° and 150° have the same terminal arm (co-terminal) and therefore have the same trigonometric ratios.



Example 5: Find three co-terminal angles of 60°

Example 6: Find the EXACT value of each of the following

a) $\sin(-45^\circ)$



b) $\cos(-60^\circ)$

